Case Study: Can We Use ML to Identify Trees Around UVA?

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Scenario:

Imagine that you are resting on the University of Virginia's lawn, enjoying the lush greenery around you. As an activist for sustainability and an immense fan of the lawn and UVA's greenery (as most UVA students and faculty are), you wonder if there exists a tool where a person can take a picture of any tree around UVA or Charlottesville and be able to find the species and information about this tree. This tool would be fascinating for its applicability to UVA. Still, it would also be a great first step towards sustainability and preserving trees, as it creates greater awareness of the trees around us, allowing us to appreciate them further. The more we know, the more likely we are to protect.

In this case study, you will be the creator of this tool and explore how computer vision can accurately classify tree images by species. You will do this using a huge dataset of real-world tree images. Then you will investigate how a Convolutional Neural Network (CNN) can accurately identify different tree species from these images. Your goal is to make this model capable of classifying new images of trees from Charlottesville!

Case Study Github: https://github.com/praggnyaKanungo/DS4002_CS3_Tree_Case_Study/tree/main

Deliverable:

This project hopes to ensure that you, as a student, can learn about image classification, exploratory data analysis with images, CNNs, and the real-world application of machine learning. Thus, your final deliverable, a GitHub Repository, should have the following:

- At least 4 visualizations for the exploratory data analysis of the initial image dataset
- A trained tree image to species classifier using CNNs
- An executed method for model accuracy evaluation
- Use of your classifier to predict the species of new tree images taken in Charlottesville
- A README.md that explains your process and results